SHoEbox: reference defaults and reviewed fixes

13 November 2026

All produced repairs for the current 18-region native design are included in the default reference path. The review also corrected primary-clock semantics and removed live semantic-ancestry eviction from the fixed eligibility pool.

Source reconciliation

GitHub main was `ea878a3`; the completion repair was on `3e5dd4d` in draft PR #2. The recovered completion package and latest audit source match that branch byte for byte. Its ancestry includes the earlier composition, multi-source binding, identity/vector, relay-preservation, contribution-check and learned-message fixes. The separate `feat/hierarchy-pilot` branch is an older six-region PyTorch study, not an additional repair to this native runtime, and remains on its own branch.

Enabled reference behavior

Mechanism	Default behavior
Constituent formation	Multi-source binding, stable identity semantics, learned completion and recursive learned exports.
Teaching and transport	Relay recognition/provenance, semantic credit, retained-input credit and credited feedback activation/routing.
State and routing	Learned bottom state, task routing, EMA prototypes and revival.
Memory	Both EC recall channels and key-trace mixing.
Auxiliary learning	DFA gain 1, separate A1 residual objective and local innovation targets.
Credit coverage	Three neural hops; all 64 spatial coordinates; every-byte, 64-coordinate bottom temporal credit; unsampled third-hop credit.

Both `fresh_model()` and low-level `Model()` select completion with this policy. `shoebox_reference.py` supplies a repository-level entry point. Explicit older kernels and saved checkpoint policies remain available for comparisons. `AGENTS.md` records that completed architectural repairs belong in the default development path and throughput regressions do not justify leaving them opt-in.

Two additional corrections

Primary clock: atoms, sequences, role bindings and relays propagate primary occurrence spans separately from coverage of all constituents. Held side context can no longer force a split at an otherwise adjacent primary boundary. The confidence-1.0 regression now joins adjacent events; genuine primary gaps and overlaps still split. `completion_clocks()` exposes this distinction.

Live semantic ancestry: a longer repeated-pattern stress exposed overwriting when the old 16,384-record pool filled with live dependencies. The pool now grows only when collection cannot free a record. Reverse traversal has no fixed work cutoff. Overflow records are serialized and restored, and stream reset clears them. This retains the saved learning graph instead of silently dropping credit.

Verification and measured cost

- 156 package tests pass, including 18 completion tests and six reference-default tests.
- Six additional default, checkpoint and migration tests pass with 64 experts per region.
- ASan/UBSan pass with enabled reference learning paths and forced pool growth, credit traversal, export/import and reset.
- A 16,416-record live dependency chain grows to 32,768 slots, retains its earliest producer, traverses every record and resumes identical updates after save/load.
- An actual checkpoint made by the unchanged version-33 source converts with exact fitted neural/semantic weights, gates and dictionaries, and a clean stream.
- The repeated-pattern smoke passes finite training, no lost semantic handles, no omitted semantic work, no live capacity pruning, no expired/duplicate forecasts, exact snapshot continuation and production/diagnostic parity.

The 1M fixture has 1,031,952 parameters. Timing uses one pinned CPU core, seed 7, 2,048 warmup targets and three interleaved rounds of 8,192 next-byte targets per runtime at learning rate 0.04. Compilation, initialization, final draining and verification are outside timed calls. The workload is repeated synthetic sentences; these are mechanism and timing measurements, not held-out language quality results.

Runtime	Median training bytes/s
Earlier actual default: composition	12,725
Earlier recommended runtime: relays	3,404
Current full reference: completion	400

The current reference is 88.2% slower than relays on this workload. Its semantic pool reached 65,536 slots. Performance optimization remains subsequent work; these features stay enabled.

Use and checkpoint contract

```
from shoebox_reference import fresh_model
model = fresh_model()
print(model.capabilities())
```

New checkpoints are **version 34 / composition ABI 8**, including variable-length semantic overflow. Version-33 completion checkpoints require explicit conversion at a fresh stream boundary. From `experiments/identity_semantics`:

```
python src/convert_to_completion.py old.npz new.npz
python -m unittest discover -s tests -v
python tests/run_completion_sanitizer.py
python src/reference_smoke.py
```

The converter accepts versions 31, 32 and 33, preserving their fitted policy. Version-31/32 checkpoints still load directly in their original kernels. Explicit `profile="historical"` retains the previous completion training settings while using the repaired clock and pool. Fresh development models use the full policy.

Remaining design work

The audit's remaining targets are reusable upper pattern classes, exported content driven by constituent-specific expert computation, local neural credit to child experts, and neural producer eligibility for long completed events. Native hippocampal recall still uses two independent EC associative transition channels; shared content/context episode identity remains to be implemented.

Eight historical experimental control families are absent from this kernel and are reported as unavailable. Three previously accepted no-op port controls now reject nonzero values. `capabilities()` reports effective switches, memory contract and semantic capacity. These absent implementations are not represented as enabled features. Source hashes and compact test/measurement files accompany the code; trained models, corpus files and compiled binaries are excluded.